

Cheng Guo

CONTACT INFORMATION

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RESEARCH INTERESTS

My research develops theoretically-grounded and computationally-scalable methods to improve the operations of large-scale markets complicated by features such as nonconvexity, stochasticity, network effects, and strategic behavior, with electricity markets as a primary motivation. On the theoretical side, I study market design with provable performance guarantees and rigorous analysis, grounded in duality theory and state-of-the-art conic programming methods. On the computational side, I develop novel decomposition and convex relaxation methods for mixed-integer nonlinear, stochastic, and robust optimization, enabling the solution of large-scale market operations problems that were previously intractable.

EXPERIENCE

Clemson University, Clemson, SC December 2021 - present
School of Mathematical and Statistical Sciences
Assistant Professor, area: Operations Research (tenure clock start: Fall 2022)

Columbia University, New York, NY 2021 - 2022
Department of Industrial Engineering and Operations Research
Visiting Researcher

- Host: Daniel Bienstock
- DOE ARPA-E PERFORM (Performance-based Energy Resource Feedback, Optimization, and Risk Management) project

EDUCATION

University of Toronto, Toronto, ON September 2017 - November 2021
Department of Mechanical and Industrial Engineering
Ph.D. in Industrial Engineering, GPA: 3.96/4.00

- Advisor: Merve Bodur
- Selected coursework: Stochastic Programming & Robust Optimization, Modeling Interactions on Networks, OM Matching Markets, Mathematical Methods in Power Systems

Columbia University, New York, NY 2015 - 2017
Department of Industrial Engineering and Operations Research
M.S. in Operations Research

- Selected coursework: Transportation Analytics & Logistics, Optimization I, Programming for Financial Engineering, Seminar on Queueing Theory

Wuhan University, Wuhan, China 2011 - 2015
School of Economics and Management
B.A. in Economics, B.S. in Mathematics

- Hongyi Honor Program Outstanding Graduates Award
- Selected coursework: Advanced Microeconomics, Industrial Organization, Advanced Financial Theory, Advanced Econometrics, Chaotic Dynamical Systems, Topology

JOURNAL
ARTICLES

- (*: corresponding author; Underline: student coauthor)
- J5. *Copositive Duality for Discrete Energy Markets* [\[pdf\]](#)
Cheng Guo*, Merve Bodur, Joshua A. Taylor
Management Science 72.3 (2026): 2022-2040
- J4. *Tightening Quadratic Convex Relaxations for the Alternating Current Optimal Transmission Switching Problem* [\[pdf\]](#)
Cheng Guo*, Harsha Nagarajan, Merve Bodur
INFORMS Journal on Computing, forthcoming
- J3. *Risk-Aware Security-Constrained Unit Commitment* [\[pdf\]](#)
Daniel Bienstock, Yury Dvorkin, Cheng Guo*, Robert Mieth, Jiayi Wang
IEEE Transactions on Energy Markets, Policy and Regulation 2.4 (2024): 536-551
- J2. *Generation Expansion Planning with Revenue Adequacy Constraints* [\[pdf\]](#)
Cheng Guo*, Merve Bodur, Dimitri J. Papageorgiou
Computers & Operations Research 142 (2022): 105736
- J1. *Logic-based Benders Decomposition and Binary Decision Diagram Based Approaches for Stochastic Distributed Operating Room Scheduling* [\[pdf\]](#)
Cheng Guo*, Merve Bodur, Dionne M. Aleman, and David R. Urbach
INFORMS Journal on Computing 33.4 (2021): 1551-1569

PREPRINTS

- P3. *Networked Markets with Production and Edge Capacity Constraints: From Competitive Equilibria To Market Entry* [\[pdf\]](#)
Cheng Guo, Jiayi Wang, Ozan Candogan
- P2. *Pricing Discrete and Nonlinear Markets With Semidefinite Relaxations* [\[pdf\]](#)
Cheng Guo, Lauren Henderson, Ryan Cory-Wright, Boshi Yang
- P1. *Incentivizing Investment and Reliability: A Study on Electricity Capacity Markets* [\[pdf\]](#)
Cheng Guo, Christian Kroer, Yury Dvorkin, Daniel Bienstock

WORKING
PAPERS

- W1. *Stochastic Dual Dynamic Programming for Multiobjective Multistage Problems*
Benjamin J. Hamlin, Cheng Guo, Margaret Wiecek

HONORS AND
AWARDS

- Finalist for student Anna Deza, INFORMS Undergraduate OR Prize Competition, 2020
- Mixed Integer Programming Workshop Student Travel Support, 2019
- Hongyi Outstanding Graduates Award, 2015
- Economics and Management School Scholarship, 2013 - 2014

FUNDING

- Clemson University - Clemson Faculty SUCCEEDS: Program 1 (Project Initiation/SEED Funding, awarded to 16 PIs university-wide) (PI), 2024
- University of Toronto - Bert Wasmund Graduate Fellowships in Sustainable Energy Research, 2018

TEACHING

Clemson University*Instructor*

- MATH 8100 - Mathematical Programming (graduate): Fall 2022, Spring 2023, Spring 2024, Fall 2024
- MATH 4400/6400 - Linear Programming (undergraduate/graduate): Fall 2024, Fall 2025, Spring 2026
- STAT 3090 - Introductory Business Statistics (undergraduate): Spring 2022 (virtual), Fall 2023, Fall 2024, Fall 2025

University of Toronto*Tutorial Teaching Assistant*

- MIE 562 - Scheduling (undergraduate/graduate): Fall 2019, Fall 2020
- MIE 335 - Algorithms and Numerical Methods (undergraduate): Winter 2019

Wuhan University*Teaching Assistant*

- Probability Theory (undergraduate): Fall 2014

ADVISING

Ph.D. Students

Lauren Henderson (since 2023), Benjamin Hamlin (since 2022, co-advised with Margaret Wiecek)

M.S. Students

Lauren Henderson (2024)

Undergraduate Students

Jiayi Wang (co-advised, Columbia B.S. 2022 → Stanford Ph.D.), Anna Deza (co-advised, U. Toronto B.A.Sc. 2020 → UC Berkeley Ph.D.), Ryan Do (co-advised, U. Toronto B.A.Sc. 2019 → U. Toronto M.Eng.)

Ph.D. Thesis Committee Member

Kristen Joyce, Sarah Kelly (2024)

M.S. Thesis Committee Member

Yunheng Jiang (2022)

INVITED TALKS

- Cornell University, FIND Seminar, Ithaca, NY April, 2024
- Mixed Integer Programming Workshop, Los Angeles, CA May, 2023
- Polytechnique Montreal, GERAD Seminar, Virtual May, 2022
- Discrete Optimization Talks, Virtual December, 2020

CONFERENCE
PRESENTATIONS

- (Upcoming) INFORMS MSOM Conference, Boston, MA July, 2026
- (Upcoming) Marketplace Innovation Workshop (MIW), Online May, 2026
- Production and Operations Management Society (POMS) Conference, Reno, NV May, 2026
- INFORMS Optimization Society Conference, Atlanta, GA March, 2026

- INFORMS Annual Meeting, Atlanta, GA October, 2025
- IEEE Power and Energy Society (PES) General Meeting, Austin, TX July, 2025
- International Conference on Continuous Optimization (ICCOPT), Los Angeles, CA July, 2025
- Production and Operations Management Society (POMS) Conference, Atlanta, GA May, 2025
- INFORMS Computing Society Conference, Toronto, ON March, 2025
- INFORMS Annual Meeting, Seattle, WA October, 2024
- International Symposium on Mathematical Programming, Montreal, QC July, 2024
- INFORMS Optimization Society Conference, Houston, TX March, 2024
- INFORMS Annual Meeting, Phoenix, AZ October, 2023
- INFORMS MSOM Conference, Montreal, QC June, 2023
- INFORMS Annual Meeting, Indianapolis, IN October 2022
- International Conference on Continuous Optimization (ICCOPT), Bethlehem, PA July, 2022
- INFORMS Optimization Society Conference, Greenville, SC March, 2022
- INFORMS Annual Meeting, Virtual October 2021
- International Conference on Game Theory (poster), Virtual July 2021
- IPCO Conference (poster), Virtual June 2021
- CORS Annual Conference, Virtual June 2021
- Mixed Integer Programming Workshop (poster), Virtual May 2021
- Grid Science Winter School (poster), Virtual January 2021
- INFORMS Annual Meeting, Virtual November 2020
- INFORMS Annual Meeting, Seattle, WA October 2019
- DIMACS Workshop on MINLP (poster), Montreal, QC October 2019
- Mixed Integer Programming Workshop (poster), Boston, MA July 2019
- Optimization Days, Montreal, QC May 2019
- INFORMS Computing Society Conference, Knoxville, TN January 2019

ACADEMIC SERVICE

- Journal reviewer for *Operations Research*, *Mathematical Programming*, *Management Science*, *Manufacturing & Service Operations Management*, *SIAM Journal on Optimization*, *INFORMS Journal on Computing*, *Transportation Science*, *Production and Operations Management*, *Computers and Operations Research*, *IEEE Transactions on Power Systems*
- Conference reviewer for *EC (2026)*, *IEEE Power & Energy Society General Meeting (2024, 2025, 2026)*
- Vice President/President-Elect for INFORMS Junior Faculty Interest Group (JFIG) (2025-2027)
- Program committee member for Mixed Integer Programming (MIP) Workshop 2026
- Session Chair for INFORMS Optimization Society Conference (2022, 2024, 2026); INFORMS Annual Meeting (2019, 2021, 2022, 2023, 2024); CORS Annual Meeting (2021)

UNIVERSITY SERVICE

- Clemson University SMSS Research Committee (2023-present)
- Co-organizer of University of Toronto MIE UTOrg Seminar (2019-2020)

ADVANCED
TRAINING

- “Cultivating an Inclusive Classroom Environment”, Academic Impressions

2024

SKILLS

- Programming language: Python, Julia, C++
- Software: Gurobi, CPLEX, Knitro, Mosek

OTHER
ACTIVITIES

- INFORMS UofT Student Chapter (Honorable Mention, 2020), Vice President (2019-2021)
- Columbia IEOR Mentorship Program, Mentor (2018-2020)
- Wuhan U. Women Soccer Team, Captain (2013-2015)